Environmental Biotechnology Laboratory

BIOT36036

Second Semester
Third Level

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Lab #6

Screening of Dye Degrading Bacteria

Part I

Isolation of dye-degraded bacteria from human hands

Isolation of dye-degraded bacteria from human hands

Introduction

Dyes are widely used in the textile, food, and cosmetic industries. They are generally resistant to conventional biological wastewater treatment systems. The release of wastewater containing dye pollutes surface water (e.g. rivers, lakes). The ability of microorganisms to decolorize and degrade dyes has been widely investigated to use for bioremediation purposes.

The presence of microorganisms on humans can be isolated by incubating agar plates after they are touched by the hands of students. The ability of microorganisms to degrade chemicals can be shown by an analytical measurement of the degradation of chemicals. When the chemicals are dyes (colorants) in water, microbial activity on degradation of dyes can be demonstrated by observing a decreasing degree of color as a result of the enzymatic activity (e.g., azoreductase).

Objective

The objective of this lab is to isolate bacteria from human hand specimens that can degrade dye in wastewater.

Materials

- Sterile nutrient agar plates.
- Metal loop.

Methods

(A) Specimens collection

(1) Bacteria are collected from the hands of volunteer students. Each student touches a Nutrient agar plates by pressing all their fingertips carefully.

(2) The plates are incubated at 37 °C for 48 hours. This plate is called an isolation plate.

(B) Obtaining pure cultures from a mixed population

(1) An isolated colony is aseptically "picked off" the isolation plate (Figure 1) and transferred to new sterile medium and incubated at 37°C for 24 hours.

(2) After incubation period, the organisms in the new culture will be descendants of the same organism, that is, a pure culture. The pure bacterial cultures are ready to use in dye degradation tests.

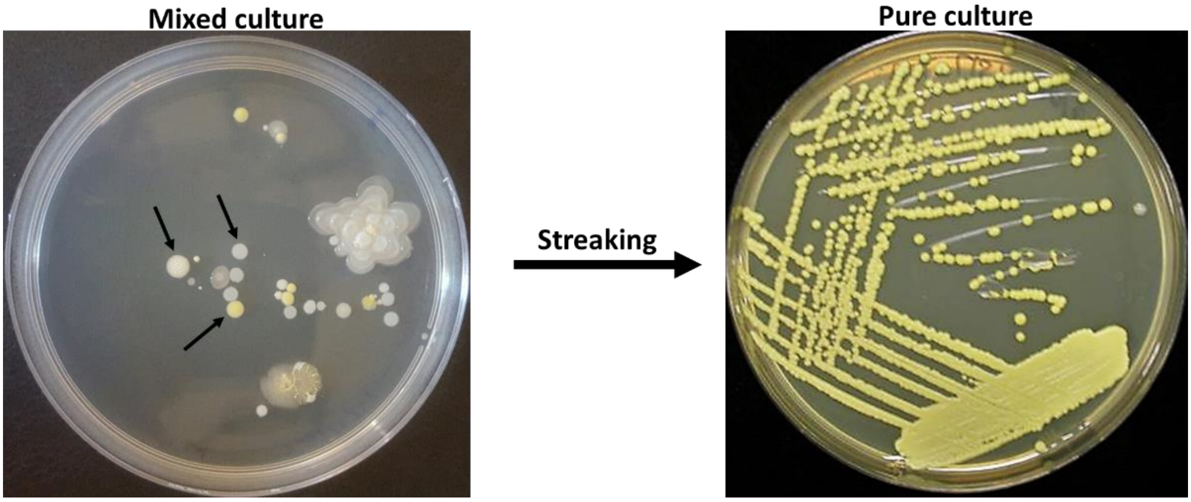

Figure : Picking a single colony off (black arrows) of isolation plate in order to obtain a pure culture.